

I would like a priority correction (e.g., a registration for the 1st occasion): ☐

Name: NEPTUN: Score: Grade:

Fundamentals of theory of computation I. – sample

0-23: grade 1, 24-31: grade 2, 32-39: grade 3, 40-47: grade 4, 48-60: grade 5

1. True or false? Write the first letter of your answer (T or F) into the box. (20x1 points)

☐ $|\{a, aa\}^2| = 4$

☐ Regular expressions $(ab)^*$ and a^*b^* describe the same language.

☐ Let V be an alphabet and $L \subseteq V^*$ be a language over V . Then $L^3 = \{uuu \mid u \in L\}$.

☐ For every homomorphism h we have $h(\varepsilon) = \varepsilon$.

☐ Every language of type 2 can be described by a regular expression.

☐ Every left linear grammar is a linear grammar.

☐ If $G = (N, \Sigma, P, S)$ is a grammar of type 3 normal form then for all the rules of P are of the form $A \rightarrow aB$ ($A, B \in N, a \in \Sigma$) or $A \rightarrow a$ ($A \in N, a \in \Sigma$).

☐ If all the rules of $G = (N, \Sigma, P, S)$ are of the form either $A \rightarrow aB$, ($A, B \in N, a \in \Sigma$) or $A \rightarrow a$ ($A \in N, a \in \Sigma$), then G is a grammar of type 3 normal form.

☐ Let $A = (Q, \Sigma, \delta, Q_0, F)$ be a nondeterministic finite automaton. Then there is a deterministic finite automaton A' with the property $L(A') = L(A)$.

☐ If L has a finite number of pairwise distinct suffix languages with respect to the words over the alphabet of L then L is a regular language.

☐ Assume that $\delta(r, a) = r$ and $\delta(r, b) = s$ hold for the deterministic finite automaton $A = (Q, \{a, b\}, \delta, q_0, F)$ for $r \in F, s \in Q$. Then $L(A, r) = \{a\}L(A, r) \cup \{b\}L(A, s)$ holds.

☐ Each reachable nonterminal of any context free grammar is an active nonterminal as well.

☐ Any grammar in Chomsky normal form is a context sensitive grammar.

☐ The class of regular languages is closed under the operation reversal.

☐ Every context sensitive grammar generates a type 0 language.

☐ Let $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$ be a pushdown automaton. Then $N(A)$ is of type 1.

☐ Let $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$ be a pushdown automaton and let $(z, q) \in \delta(z, q, b)$ be a transition of A ($z \in Z, b \in \Sigma, q \in Q$). Applying this transition the stack will contain one more letter than in the previous configuration.

☐ Let A be a pushdown automaton. Then there is a deterministic pushdown automaton A' with the property $L(A') = L(A)$.

☐ Let $G = (N, \Sigma, P, S)$ be a context free grammar. If a node labelled by A in a derivation tree of a word in $L(G)$ has two children, the left child being labelled by B , the right child being labelled by C then $A \rightarrow BC \in P$.

☐ Every grammar in Kuroda normal form is a length-nondecreasing grammar.

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2. Answer the following questions. No explanation is required.

(10x2 points)

- (a) $\emptyset^0 =$
- (b) Let $G = (N, \Sigma, P, S)$ be a grammar és $u, v \in (N \cup \Sigma)^*$. Define what does it mean that v can be derived from u in one step. ($u \Rightarrow_G v$ relation)
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- (c) Let $G_1 = (\{A\}, \{a, b\}, \{A \rightarrow abA \mid b\}, A)$ and $G_2 = (\{B\}, \{a, b\}, \{B \rightarrow baB \mid b\}, B)$ be regular grammars. Give the a grammar of type 3 generating $L(G_1)L(G_2)$ according to the construction we have learnt. (Don't forget to name the initial symbol.)
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- (d) Let $A = (Q, \Sigma, \delta, Q_0, F)$ be a nondeterministic finite automaton. Give the definition of the transition function δ .
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- (e) Consider a nondeterministic finite automaton A with 4 states and 3 accepting states. How many accepting states are there in the deterministic finite automaton A' we have learnt to be equivalent with A ? (The set of states of A' is the powerset of the set of states of A .)
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- (f) Consider the following CF grammar. (S is the initial symbol, a, b are the terminals).
 $S \rightarrow SA \mid CX, A \rightarrow \epsilon \mid XY, B \rightarrow AA, C \rightarrow AX, X \rightarrow a, Y \rightarrow b$
In the algorithm of bringing this grammar to a Chomsky normal form what will be set U in the 'removing ϵ rules' step?
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- (g) Let q be a state of a deterministic finite automaton A . What condition should hold for q if $L(A, q)$ contains ϵ ?
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- (h) Give a pushdown automaton rule with the following effect. By an ϵ -move the automaton removes a letter b from the pushdown store and changes the state of the automaton from q to r .
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- (i) Let $G = (N, \Sigma, P, S)$ be a context free grammar. Define what is meant by an active nonterminal of G .
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- (j) Let $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$ be a pushdown automaton. Give the definition $N(A)$, the language accepted by A by empty store.
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NEPTUN:

3.

(4+8+4+4 points)

- (a) Let $G = (N, \Sigma, P, S)$ be a grammar of type 3. Give the construction of a grammar of type 3 generating $L(G)^*$ according to the construction we have learnt.
- (b) Define what is the membership problem of context free grammars. Introduce the algorithmic solution of Cocke, Younger and Kasami (CYK algorithm). What can you say about the efficiency of the algorithm?
- (c) Define the one-step reduction relation of a pushdown automaton $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$. ($\Rightarrow_A$ relation)
- (d) State Bar Hillel Lemma (pumping lemma for CF languages).